

Proposition A.1. For fixed $0 < \delta < \frac{1}{4}$, we have

$$\limsup_{n \rightarrow \infty} n^{-\delta} \log P_n \left(\inf_{t \geq 0} \left| \frac{1}{\sqrt{n}} \sum_{k \leq t\sqrt{n}} X_k - L(t) \right| > n^{-\frac{1}{4} + \delta} \right) < 0.$$

Remark. In particular, we have

$$\lim_{n \rightarrow \infty} P_n \left(\sup_{t \geq 0} \left| \frac{1}{\sqrt{n}} \sum_{k \leq t\sqrt{n}} X_k - L(t) \right| \leq n^{-\frac{1}{4} + \delta} \right) = 1.$$

Proof. Let $d(n)$ be the number of distinct parts partitions of n . We require only a weak form of the well-known asymptotic expansion of $d(n)$ [11],

$$d(n) = e^{\frac{2\sqrt{n}}{A} + O(\log n)}.$$

Let $a_n, b_n \in \mathbb{N}_0$ and define α_n, β_n by

$$a_n = \alpha_n \sqrt{n}, \quad b_n = \beta_n \sqrt{n}.$$

Assume that $\alpha_n, \beta_n \geq n^{-\frac{1}{4} + \delta}$. We use the saddle point bound to write, for any $x_n \in \mathbb{R}$,

$$\begin{aligned} P_n \left(\frac{1}{\sqrt{n}} \sum_{k \leq a_n} X_k = b_n \right) &= \frac{1}{d(n)} [q^n] [\zeta^{b_n}] \prod_{k \leq a_n} (1 + \zeta q^k) \prod_{k > a_n} (1 + q^k) \\ &\leq \frac{1}{d(n)} q_n^{-n} e^{-b_n x_n} \prod_{k \leq a_n} (1 + e^{x_n} q_n^k) \prod_{k > a_n} (1 + q_n^k) \\ &= \exp \left(\frac{\sqrt{n}}{A} - \log d(n) - \beta_n x_n \sqrt{n} + \sum_{k \leq a_n} \log \left(1 + e^{x_n - \frac{k}{A\sqrt{n}}} \right) + \sum_{k > a_n} \log \left(1 + e^{-\frac{k}{A\sqrt{n}}} \right) \right). \end{aligned}$$

We will take $x_n \in \{\pm n^{-\frac{1}{4}}\}$, where the sign will depend on b_n . By Taylor's Theorem,

$$\begin{aligned} \left| \log \left(1 + e^{x_n - \frac{k}{A\sqrt{n}}} \right) - \log \left(1 + e^{-\frac{k}{A\sqrt{n}}} \right) - x_n \frac{e^{-\frac{k}{A\sqrt{n}}}}{1 + e^{-\frac{k}{A\sqrt{n}}}} \right| &\leq \frac{x_n^2}{2} \sup_{|x| \leq n^{\frac{1}{4}}} \frac{e^{x - \frac{k}{A\sqrt{n}}}}{\left(1 + e^{x - \frac{k}{A\sqrt{n}}} \right)^2} \\ &\leq x_n^2 \frac{e^{-\frac{k}{A\sqrt{n}}}}{\left(1 + \frac{1}{2} e^{-\frac{k}{A\sqrt{n}}} \right)^2}. \end{aligned}$$

Thus,

$$\begin{aligned} P_n \left(\frac{1}{\sqrt{n}} \sum_{k \leq a_n} X_k = b_n \right) &\leq \exp \left(\frac{\sqrt{n}}{A} - \log d(n) + \sum_{k \geq 1} \log \left(1 + e^{-\frac{k}{A\sqrt{n}}} \right) - \beta_n x_n \sqrt{n} \right. \\ &\quad \left. + x_n \sum_{k \leq a_n} \frac{e^{-\frac{k}{A\sqrt{n}}}}{1 + e^{-\frac{k}{A\sqrt{n}}}} + O \left(x_n^2 \sum_{k \leq a_n} \frac{e^{-\frac{k}{A\sqrt{n}}}}{\left(1 + \frac{1}{2} e^{-\frac{k}{A\sqrt{n}}} \right)^2} \right) \right). \end{aligned}$$